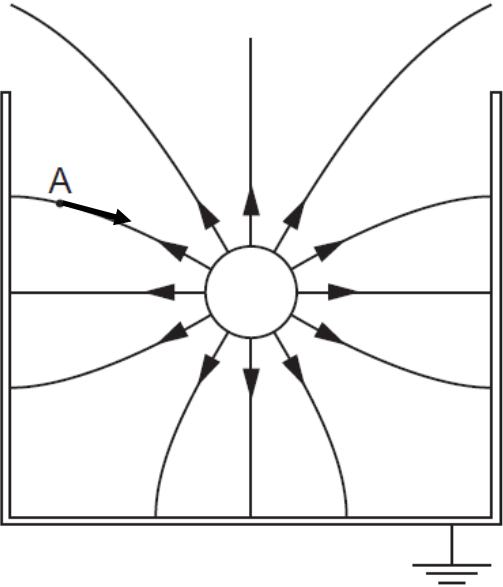


7a	Electric field strength is defined as the <u>electric</u> force <u>per</u> unit <u>positive</u> charge placed at a point in the electric field.	B2
7b(i)1	The sphere is <u>positively</u> charged because the sphere has a higher potential than the metal box which is earthed, at 0 V, based on the direction of the field lines. OR field lines pointing/directed away from the sphere	A0 M1
7b(i)2	The field lines are normal to the surface OR the lines appear to radiate from the centre of the sphere.	B1

7b(ii)	 Direction of the force is correct and tangent to the field and, origin at A.	B1
7biii1	$V = \frac{0.76 \times 10^{-9}}{4\pi\epsilon(2.4 \times 10^{-2})}$ $= 2.847 \times 10^2 \text{ V}$ $\approx 2.85 \times 10^2 \text{ V (OR } 2.8 \times 10^2 \text{ V for 2sf)}$	C1 A1
7biii2	The sign is negative. The actual potential at the surface of the sphere is smaller because the actual potential is the scalar sum of the positive potential due to the charged metal sphere and the negative potentials due to the metal box.	B1 A1 M1
7b(iv)	Electric force $F = \frac{Q_1 Q_2}{4\pi\epsilon_0 r^2}$ $= \frac{(1.60 \times 10^{-19})^2}{4\pi\epsilon_0 (6.0 \times 10^{-10})^2} = 6.39 \times 10^{-10} \text{ N}$ Ratio = $\frac{6.39 \times 10^{-10}}{9.11 \times 10^{-31} \times 9.81}$ $= 7.15 \times 10^{19}$	C1 A1
7b(v)	The gravitational forces are always attractive. OR gravitational field lines must be directed towards both box and sphere.	B1
7c(i)	As seen from <u>the increased balance reading</u>, there is a <u>downward force on magnet</u> due to wire carrying current. By Newton's third law, there is an <u>upward force on wire</u> by magnet. By Fleming's left hand rule, pole P is a north pole.	M1 M1 A1
7c(ii)	By Newton's 2nd law, $W - BIL = 0 \Rightarrow W = BIL$ $2.3 \times 10^{-3} \times 9.81 = B \times 2.6 \times 4.4 \times 10^{-2}$ $B = 0.20 \text{ T (g = 10, loses this mark)}$	C1 A1
7c(iii)	The reading will vary between 3.3 g above the original value (when there is no current) and 3.3 g below the original value. For marking: Correct peak value – 1m. Description of varying reading above and below the original value – 1m.	B2

8a	The value of V when $I = 0$ remains unchanged as intensity is varied	R1
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